量子力学习题解答
第二章　波函数与 Schrödinger 方程

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教材：周世勋《量子力学教程》第二版

2.1

几率流密度为

$$\mathbf{j}=-\frac{\mathrm{i}\hbar}{2m}(\psi^*(\mathbf{r},t)\nabla\psi(\mathbf{r},t)-\psi(\mathbf{r},t)\nabla\psi^*(\mathbf{r},t)), \tag{1}$$

设

$$\psi(\mathbf{r},t)=\phi(\mathbf{r})e^{-\mathrm{i}\omega t},$$

则

$$\begin{aligned}\mathbf{j}&=-\frac{\mathrm{i}\hbar}{2m}([\phi(\mathbf{r})e^{-\mathrm{i}\omega t}]^*\nabla\phi(\mathbf{r})e^{-\mathrm{i}\omega t}-\phi(\mathbf{r})e^{-\mathrm{i}\omega t}[\nabla\phi(\mathbf{r})(e^{-\mathrm{i}\omega t}]^*)\\&=-\frac{\mathrm{i}\hbar}{2m}[\phi^*(\mathbf{r})e^{\mathrm{i}\omega t}\nabla\phi(\mathbf{r})e^{-\mathrm{i}\omega t}-\phi(\mathbf{r})e^{-\mathrm{i}\omega t}e^{\mathrm{i}\omega t}\nabla\phi^*(\mathbf{r})]\\&=-\frac{\mathrm{i}\hbar}{2m}[\phi^*(\mathbf{r})\nabla\phi(\mathbf{r})-\phi(\mathbf{r})\nabla\phi^*(\mathbf{r})],\end{aligned} \tag{2}$$

故

$$\frac{\partial\mathbf{j}}{\partial t}=0. \tag{3}$$

即定态中几率流密度 $\mathbf{j}$ 与时间 t 无关.

2.2

由 (2) 式易得

$$\begin{aligned}\mathbf{j}&=-\frac{\mathrm{i}\hbar}{2m}[\phi_1^*(\mathbf{r})\nabla\phi_1(\mathbf{r})-\phi_1(\mathbf{r})\nabla\phi_1^*(\mathbf{r})]=-\frac{\mathrm{i}\hbar}{2m}[(\frac{1}{r}e^{\mathrm{i}kr})^*\nabla(\frac{1}{r}e^{\mathrm{i}kr})-\frac{1}{r}e^{\mathrm{i}kr}\nabla(\frac{1}{r}e^{\mathrm{i}kr})^*]\\&=-\frac{\mathrm{i}\hbar}{2m}[\frac{1}{r}e^{-\mathrm{i}kr}\nabla(\frac{1}{r}e^{\mathrm{i}kr})-\frac{1}{r}e^{\mathrm{i}kr}\nabla(\frac{1}{r}e^{-\mathrm{i}kr})]\\&=-\frac{\mathrm{i}\hbar}{2m}(\frac{1}{r}e^{-\mathrm{i}kr}[e^{\mathrm{i}kr}\nabla\frac{1}{r}+\frac{1}{r}\nabla e^{\mathrm{i}kr}]-\frac{1}{r}e^{\mathrm{i}kr}[e^{-\mathrm{i}kr}\nabla\frac{1}{r}+\frac{1}{r}\nabla e^{-\mathrm{i}kr}])\\&=-\frac{\mathrm{i}\hbar}{2m}(\frac{1}{r}e^{-\mathrm{i}kr}[-\frac{\mathbf{r}}{r^3}e^{\mathrm{i}kr}+\frac{1}{r}\nabla e^{\mathrm{i}kr}]-\frac{1}{r}e^{\mathrm{i}kr}[-\frac{\mathbf{r}}{r^3}e^{-\mathrm{i}kr}+\frac{1}{r}\nabla e^{-\mathrm{i}kr}])\end{aligned}$$

$$= -\frac{\mathrm{i}\hbar}{2m}[-\frac{\mathbf{r}}{r^4} + \frac{\exp(-\mathrm{i}kr)}{r^2}\nabla e^{\mathrm{i}kr} + \frac{\mathbf{r}}{r^4} - \frac{\exp(\mathrm{i}kr)}{r^2}\nabla e^{-\mathrm{i}kr}]$$
$$= -\frac{\mathrm{i}\hbar}{2m}[\frac{\exp(-\mathrm{i}kr)}{r^2}\nabla e^{\mathrm{i}kr} - \frac{\exp(\mathrm{i}kr)}{r^2}\nabla e^{-\mathrm{i}kr}]. \tag{4}$$

下面单独计算 $\nabla e^{\mathrm{i}kr}$:

$$\begin{aligned}
\nabla e^{\mathrm{i}kr} &= \nabla \exp(\mathrm{i}k\sqrt{x^2+y^2+z^2}) \\
&= \frac{\partial}{\partial x}\exp(\mathrm{i}k\sqrt{x^2+y^2+z^2})\mathbf{i} + \frac{\partial}{\partial y}\exp(\mathrm{i}k\sqrt{x^2+y^2+z^2})\mathbf{j} + \frac{\partial}{\partial z}\exp(\mathrm{i}k\sqrt{x^2+y^2+z^2})\mathbf{k} \\
&= \frac{2x}{2\sqrt{x^2+y^2+z^2}}\mathrm{i}k\exp(\mathrm{i}k\sqrt{x^2+y^2+z^2})\mathbf{i} + \frac{2y}{2\sqrt{x^2+y^2+z^2}}\mathrm{i}k\exp(\mathrm{i}k\sqrt{x^2+y^2+z^2})\mathbf{j} \\
&+ \frac{2z}{2\sqrt{x^2+y^2+z^2}}\mathrm{i}k\exp(\mathrm{i}k\sqrt{x^2+y^2+z^2})\mathbf{k} \\
&= \frac{\mathrm{i}kx}{r}e^{\mathrm{i}kr}\mathbf{i} + \frac{\mathrm{i}ky}{r}e^{\mathrm{i}kr}\mathbf{j} + \frac{\mathrm{i}kz}{r}e^{\mathrm{i}kr}\mathbf{k} \\
&= \frac{\mathrm{i}k}{r}e^{\mathrm{i}kr}(x\mathbf{i}+y\mathbf{j}+z\mathbf{k}) = \mathrm{i}ke^{\mathrm{i}kr}\frac{\mathbf{r}}{r}.
\end{aligned} \tag{5}$$

同理,

$$\nabla e^{-\mathrm{i}kr} = -\mathrm{i}ke^{-\mathrm{i}kr}\frac{\mathbf{r}}{r}. \tag{6}$$

代入 (4) 式可知:

$$\begin{aligned}
\mathbf{j} &= -\frac{\mathrm{i}\hbar}{2m}[\frac{\exp(-\mathrm{i}kr)}{r^2}\nabla\exp(\mathrm{i}kr) - \frac{\exp(\mathrm{i}kr)}{r^2}\nabla\exp(-\mathrm{i}kr)] \\
&= -\frac{\mathrm{i}\hbar}{2m}[\frac{\exp(-\mathrm{i}kr)}{r^2}\mathrm{i}ke^{\mathrm{i}kr}\frac{\mathbf{r}}{r} - \frac{\exp(\mathrm{i}kr)}{r^2}(-\mathrm{i}ke^{-\mathrm{i}kr}\frac{\mathbf{r}}{r})] \\
&= -\frac{\mathrm{i}\hbar}{2m}(2\mathrm{i}k\frac{\mathbf{r}}{r^3}) = \frac{\hbar k}{mr^3}\mathbf{r}.
\end{aligned} \tag{7}$$

这说明 ψ_1 是一个爆炸球面波；类似地，我们能得出 ψ_2 是一个汇聚球面波，其几率流密度为:

$$\mathbf{j} = -\frac{\hbar k}{mr^3}\mathbf{r}. \tag{8}$$

2.3

列出 Schrödinger 方程

$$\mathrm{i}\hbar\frac{\partial}{\partial t}\psi(x,t) = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi(x,t) + V(x)\psi(x,t), \tag{9}$$

分离变量：令 $\psi(x,t) = \phi(x)e^{-\mathrm{i}\omega t}$，则

$$\mathrm{i}\hbar\frac{\partial}{\partial t}(\phi(x)e^{-\mathrm{i}\omega t}) = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}(\phi(x)e^{-\mathrm{i}\omega t}) + V(x)\phi(x)e^{-\mathrm{i}\omega t}, \tag{10}$$

各自对时间和位置求偏微商后得

$$\mathrm{i}\hbar\phi(x)(-\mathrm{i}\omega)e^{-\mathrm{i}\omega t} = -\frac{\hbar^2}{2m}e^{-\mathrm{i}\omega t}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\phi(x) + V(x)\phi(x)e^{-\mathrm{i}\omega t}, \tag{11}$$

或者

$$\mathrm{i}\hbar\phi(x)(-\mathrm{i}\omega) = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\phi(x) + V(x)\phi(x), \tag{12}$$

移项整理:

$$\hbar\omega\phi(x) = E\phi(x) = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\phi(x) + V(x)\phi(x). \tag{13}$$

在 $x < 0$ 和 $x > a$ 处受无穷深方势阱影响，$\phi(x) = 0$，而在 $0 \leq x \leq a$ 内有

$$E\phi(x) = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\phi(x). \tag{14}$$

这是一个二阶常微分方程. 为此，我们设 $\phi(x) = e^{\lambda x}$，则

$$\phi^{'}(x) = \lambda e^{\lambda x}, \quad \phi^{''}(x) = \lambda^2 e^{\lambda x}, \tag{15}$$

代入 (14) 式

$$Ee^{\lambda x} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}e^{\lambda x}, \tag{16}$$

求微商后并化简可得

$$\lambda^2 = -\frac{2mE}{\hbar^2} \qquad \lambda = \pm\sqrt{-\frac{2mE}{\hbar^2}}, \tag{17}$$

或者

$$\lambda = \pm\mathbf{i}\sqrt{\frac{2mE}{\hbar^2}}. \tag{18}$$

所以我们得到 $\phi(x)$ 的两个解:

$$\phi_1(x) = \exp(\mathbf{i}\sqrt{\frac{2mE}{\hbar^2}}x), \qquad \phi_2(x) = \exp(-\mathbf{i}\sqrt{\frac{2mE}{\hbar^2}}x). \tag{19}$$

考虑到边条件 $\phi(0) = \phi(a) = 0$，而上式并不满足，故取 ϕ_1 和 ϕ_2 的一个线性组合 $\phi(x) = C_1\phi_1(x) + C_2\phi_2(x)$，使得 $\phi(0) = 0$，即

$$C_1\exp(\mathbf{i}\sqrt{\frac{2mE}{\hbar^2}}0) + C_2\exp(-\mathbf{i}\sqrt{\frac{2mE}{\hbar^2}}0) = C_1 + C_2 = 0, \tag{20}$$

易得 $C_1 = -C_2$, 故

$$\begin{aligned}
\phi(x) &= C_1\phi_1(x) - C_1\phi_2(x) = C_1[\phi_1(x) - \phi_2(x)] \\
&= C_1[\exp(\mathbf{i}\sqrt{\frac{2mE}{\hbar^2}}x) - \exp(-\mathbf{i}\sqrt{\frac{2mE}{\hbar^2}}x)] \\
&= 2\mathrm{i}C_1\frac{\exp(\mathbf{i}\sqrt{\frac{2mE}{\hbar^2}}x) - \exp(-\mathbf{i}\sqrt{\frac{2mE}{\hbar^2}}x)}{2\mathrm{i}} \\
&\equiv D\sin(\sqrt{\frac{2mE}{\hbar^2}}x),
\end{aligned} \tag{21}$$

再令 $\phi(a) = 0$, 则

$$\phi(a) = D\sin(\sqrt{\frac{2mE}{\hbar^2}}a) = 0, \qquad \sqrt{\frac{2mE}{\hbar^2}}a = n\pi, \quad (n = 1, 2, 3...). \tag{22}$$

得到了能级和含待定系数的波函数 (不妨给它们加上下角标 n)

$$E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}, \qquad \phi_n(x) = D_n \sin(\frac{n\pi}{a}x). \tag{23}$$

最后归一化（注意到 $\phi_n(x)$ 是实的，所以取模平方时可直接将双竖线去掉)：

$$\begin{aligned} 1 &= \int_{-\infty}^{+\infty} |\phi_n(x)|^2 \,\mathrm{d}x = \int_0^a |\phi_n(x)|^2 \,\mathrm{d}x \\ &= \int_0^a D_n^2 \sin^2(\frac{n\pi}{a}x)\mathrm{d}x = D_n^2 \int_0^a \frac{1-\cos(\frac{2n\pi}{a}x)}{2}\mathrm{d}x = \frac{D_n^2 a}{2}, \end{aligned} \tag{24}$$

$$D_n^2 = \frac{2}{a}, \tag{25}$$

不妨取 $D_n = \sqrt{\frac{2}{a}}$, 则我们得到最后的波函数

$$\phi_n(x) = \sqrt{\frac{2}{a}} \sin(\frac{n\pi}{a}x). \qquad (n = 1, 2, 3...) \tag{26}$$

2.4 证明

波函数为

$$\psi_n = \begin{cases} A \sin \dfrac{n\pi}{2a}(x+a) \quad (n = 1, 2, 3...) & |x| < 0, \\ 0 & |x| \geq a. \end{cases} \tag{27}$$

归一化：

$$\begin{aligned} 1 &= \int_{-\infty}^{+\infty} \psi_n^2(x)\mathrm{d}x = \int_{-a}^{a} \psi_n^2(x)\mathrm{d}x = \int_{-a}^{a} A_n^2 \sin^2[\frac{n\pi}{2a}(x+a)]\mathrm{d}x \\ &= A_n^2 \int_{-a}^{a} \frac{1-\cos[\frac{n\pi}{a}(x+a)]}{2}\mathrm{d}x = A_n^2 a, \end{aligned} \tag{28}$$

$$A_n^2 = \frac{1}{a}, \tag{29}$$

不妨取 $A_n = \sqrt{\frac{1}{a}}$.

2.5

一维谐振子的归一化波函数满足

$$\phi_n(x) = (\frac{\sqrt{m\omega_0/\hbar}}{\sqrt{\pi}2^n n!})^{\frac{1}{2}} \exp(-\frac{1}{2}\frac{m\omega_0}{\hbar}x^2) H_n(\sqrt{\frac{m\omega_0}{\hbar}}x), \tag{30}$$

当 $n = 1$ 时，

$$\begin{aligned} \phi_1(x) &= (\frac{\sqrt{m\omega_0/\hbar}}{\sqrt{\pi}2})^{\frac{1}{2}} \exp(-\frac{1}{2}\frac{m\omega_0}{\hbar}x^2) H_1(\sqrt{\frac{m\omega_0}{\hbar}}x) = \sqrt{2}\frac{(m\omega_0)^{\frac{1}{4}}}{(\pi\hbar)^{\frac{1}{4}}}\frac{(m\omega_0)^{\frac{1}{2}}}{\hbar^{\frac{1}{2}}} \exp(-\frac{1}{2}\frac{m\omega_0}{\hbar}x^2)x \\ &= (\frac{2}{\sqrt{\pi}})^{\frac{1}{2}}(\frac{m\omega_0}{\hbar})^{\frac{3}{4}} \exp(-\frac{1}{2}\frac{m\omega_0}{\hbar}x^2)x, \end{aligned} \tag{31}$$

$$|\phi_1(x)|^2 = \frac{2}{\sqrt{\pi}}(\frac{m\omega_0}{\hbar})^{\frac{3}{2}} \exp(-\frac{m\omega_0}{\hbar}x^2)x^2, \tag{32}$$

令

$$\frac{\mathrm{d}}{\mathrm{d}x}|\phi_1(x)|^2 = 0, \tag{33}$$

易知密度函数的驻点为 $x = 0$ 和 $x = \pm\sqrt{\frac{\hbar}{m\omega_0}}$，容易验证当 x 取后者时，几率密度函数有最大值.

2.6 证明

由于 $V(x) = V(-x)$, 可设

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\psi(x) + V(x)\psi(x) = E\psi(x), \tag{34}$$

取 $x' = -x$, 则

$$\begin{aligned} -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\psi(-x) + V(x)\psi(-x) &= -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\psi(x') + V(-x')\psi(x') \\ &= -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\psi(x') + V(x')\psi(x') \\ &= E\psi(-x) = E\psi(x'), \end{aligned} \tag{35}$$

故 $\psi(x)$ 与 $\psi(-x)$ 均为方程的解.

当 E 非简并（或者说 $\psi(x)$ 与 $\psi(-x)$ 线性相关）时, 可设 $\psi(-x) = C\psi(x)$, 则

$$\psi(x) = \psi(-(-x)) = C\psi(-x) = C \cdot C\psi(x) = C^2\psi(x), \tag{36}$$

$$C^2 = 1 \qquad C = \pm 1. \tag{37}$$

若 $C = 1$，则 $\psi(x) = \psi(-x)$，为偶宇称；若 $C = -1$，则 $-\psi(x) = \psi(-x)$，为奇宇称.

当 E 简并（或者说 $\psi(x)$ 与 $\psi(-x)$ 线性无关）时, 可设

$$\psi_1(x) = \psi(x) + \psi(-x) \qquad \psi_2(x) = \psi(x) - \psi(-x), \tag{38}$$

则

$$\psi_1(-x) = \psi(-x) + \psi(x) = \psi_1(x) \tag{39}$$

为偶宇称;

$$\psi_2(-x) = \psi(-x) - \psi(x) = -\left[\psi(x) - \psi(-x)\right] = -\psi_2(x) \tag{40}$$

为奇宇称.

2.7 有限深对称方势阱问题

体系的 Schrödinger 方程:

当 $|x| \leq a$ 时

$$E\psi(x) = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\psi(x). \tag{41}$$

当 $|x| > a$ 时

$$E\psi(x) = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\psi(x) + V(x)\psi(x). \tag{42}$$

在 $|x| \leq a$ 内, 设 $\psi(x) = e^{\lambda x}$, 则

$$\psi^{'}(x) = \lambda e^{\lambda x}, \quad \psi^{''}(x) = \lambda^2 e^{\lambda x}, \tag{43}$$

代入 (41) 式

$$Ee^{\lambda x} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}e^{\lambda x}, \tag{44}$$

求微商后并化简可得

$$\lambda^2 = -\frac{2mE}{\hbar^2} \qquad \lambda = \pm\sqrt{-\frac{2mE}{\hbar^2}} = \pm\mathrm{i}\sqrt{\frac{2mE}{\hbar^2}} \equiv \pm\mathrm{i}k. \tag{45}$$

所以我们取 $\psi(x)$ 的一个线性组合:

$$\psi(x) = C_1 e^{\mathrm{i}kx} + C_2 e^{-\mathrm{i}kx}. \tag{46}$$

在 $|x| > a$ 内, 设 $\psi(x) = e^{\alpha x}$, 则

$$\psi^{'}(x) = \alpha e^{\alpha x}, \quad \psi^{''}(x) = \alpha^2 e^{\alpha x}, \tag{47}$$

代入 (42) 式

$$(E - V_0)e^{\alpha x} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}e^{\alpha x}, \tag{48}$$

求微商后并化简可得

$$\alpha^2 = -\frac{2m(E - V_0)}{\hbar^2} \qquad \alpha = \pm\sqrt{-\frac{2m(E - V_0)}{\hbar^2}} \pm\sqrt{\frac{2m(V_0 - E)}{\hbar^2}} \equiv \pm\beta. \tag{49}$$

注意到当 x 趋向 ∞ 时, $\psi(x)$ 趋向 0, 所以我们取

$$\psi(x) = \begin{cases} Ae^{-\beta x} & x \geq a, \\ Be^{\beta x} & x \leq -a. \end{cases} \tag{50}$$

我们要求 $\psi(x)$ 及其一阶微商在边界处连续, 所以

当 $\psi(a) = 0$ 时,

$$\begin{aligned} Ae^{-\beta a} &= C_1 e^{\mathrm{i}ka} + C_2 e^{-\mathrm{i}ka} \\ -\beta Ae^{-\beta a} &= \mathrm{i}kC_1 e^{\mathrm{i}ka} - \mathrm{i}kC_2 e^{-\mathrm{i}ka}. \end{aligned} \tag{51}$$

两式相除得

$$-\beta = \mathrm{i}k\frac{C_1 e^{\mathrm{i}ka} - C_2 e^{-\mathrm{i}ka}}{C_1 e^{\mathrm{i}ka} + C_2 e^{-\mathrm{i}ka}}. \tag{52}$$

当 $\psi(-a) = 0$ 时,

$$\begin{aligned} Be^{\beta(-a)} &= C_1 e^{\mathrm{i}k(-a)} + C_2 e^{-\mathrm{i}k(-a)} \\ \beta Be^{\beta(-a)} &= \mathrm{i}kC_1 e^{\mathrm{i}k(-a)} - \mathrm{i}kC_2 e^{-\mathrm{i}k(-a)}. \end{aligned} \tag{53}$$

两式相除得

$$\beta = \mathrm{i}k\frac{C_1 e^{\mathrm{i}k(-a)} - C_2 e^{\mathrm{i}ka}}{C_1 e^{\mathrm{i}k(-a)} + C_2 e^{\mathrm{i}ka}}. \tag{54}$$

由 (52) 式和 (54) 式联立可知

$$C_1 = \pm C_2. \tag{55}$$

另外，考虑到

$$\cos\theta = \frac{e^{\mathrm{i}\theta} + e^{-\mathrm{i}\theta}}{2}, \qquad \sin\theta = \frac{e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta}}{2\mathrm{i}}. \tag{56}$$

易得

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = -\mathrm{i}\frac{e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta}}{e^{\mathrm{i}\theta} + e^{-\mathrm{i}\theta}}. \tag{57}$$

$$\cot\theta = \frac{\cos\theta}{\sin\theta} = \mathrm{i}\frac{e^{\mathrm{i}\theta} + e^{-\mathrm{i}\theta}}{e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta}}. \tag{58}$$

上述 θ 均为任意角度. 接下来我们讨论 (55) 式.

当 $C_1 = C_2$ 时, 可知

$$\beta = -\mathrm{i}k\frac{e^{\mathrm{i}ka} - e^{-\mathrm{i}ka}}{e^{\mathrm{i}ka} + e^{-\mathrm{i}ka}} = k\tan ka. \tag{59}$$

这里我们在最后一步用到了 (57) 式. 我们令 (59) 式两边同乘 a，则

$$\beta a = ka\tan ka. \tag{60}$$

再令 $\beta a = \eta$，$ka = \xi$，则

$$\eta = \xi\tan\xi. \tag{61}$$

当 $C_1 = -C_2$ 时, 可知

$$\beta = \mathrm{i}k\frac{e^{-\mathrm{i}ka} + e^{\mathrm{i}ka}}{e^{-\mathrm{i}ka} - e^{\mathrm{i}ka}} = -k\cot ka. \tag{62}$$

这里我们在最后一步用到了 (58) 式. 我们令 (62) 式两边同乘 a，则

$$\beta a = -ka\cot ka. \tag{63}$$

再令 $\beta a = \eta$，$ka = \xi$，则

$$\eta = -\xi\cot\xi. \tag{64}$$

综上所述，(61) 式和 (64) 式即为所求.

2.8

体系满足 Schrödinger 方程

$$\psi(x) = \begin{cases} \psi_1 = 0, & x \le 0, \\ \psi_2(x), & 0 \le x \le a, \\ \psi_3(x), & a \le x \le b, \\ \psi_4(x), & x \ge b. \end{cases} \tag{65}$$

继而写成

$$\begin{cases}(E-V_0)\psi_2(x)=-\dfrac{\hbar^2}{2m}\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}\psi_2(x), & 0\le x\le a,\\ (E+V_1)\psi_3(x)=-\dfrac{\hbar^2}{2m}\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}\psi_3(x), & a\le x\le b,\\ E\psi_4(x)=-\dfrac{\hbar^2}{2m}\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}\psi_4(x), & x\ge b.\end{cases}\tag{66}$$

因此，仿照之前的做法，我们设 $\psi_2(x)=e^{\lambda_1 x}$, 则 $\psi_2'(x)=\lambda_1 e^{\lambda_1 x}$, $\psi_2''(x)=\lambda_1^2 e^{\lambda_1 x}$.

$$(E-V_0)e^{\lambda_1 x}=-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}e^{\lambda_1 x},\tag{67}$$

求微商后并化简可得

$$\lambda_1^2=\frac{2m(V_0-E)}{\hbar^2}\qquad \lambda_1=\pm\sqrt{\frac{2m(V_0-E)}{\hbar^2}}\equiv\pm\beta.\tag{68}$$

所以让我们取线性组合

$$\psi_2(x)=C_1e^{\beta x}+C_2e^{-\beta x}.\tag{69}$$

设 $\psi_3(x)=e^{\lambda_2 x}$, 则 $\psi_3'(x)=\lambda_2 e^{\lambda_2 x}$, $\psi_3''(x)=\lambda_2^2 e^{\lambda_2 x}$.

$$(E+V_1)e^{\lambda_2 x}=-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}e^{\lambda_2 x},\tag{70}$$

求微商后并化简可得

$$\lambda_2^2=-\frac{2m(E+V_1)}{\hbar^2}\qquad \lambda_2=\pm\mathrm{i}\sqrt{\frac{2m(E+V_1)}{\hbar^2}}\equiv\pm\mathrm{i}k.\tag{71}$$

所以让我们取线性组合

$$\psi_3(x)=D_1e^{\mathrm{i}kx}+D_2e^{-\mathrm{i}kx}.\tag{72}$$

设 $\psi_4(x)=e^{\lambda_3 x}$, 则 $\psi_4'(x)=\lambda_3 e^{\lambda_3 x}$, $\psi_4''(x)=\lambda_3^2 e^{\lambda_3 x}$.

$$Ee^{\lambda_3 x}=-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}e^{\lambda_3 x},\tag{73}$$

求微商后并化简可得

$$\lambda_3^2=-\frac{2mE}{\hbar^2}\qquad \lambda_3=\pm\mathrm{i}\sqrt{\frac{2mE}{\hbar^2}}\equiv\pm\mathrm{i}\alpha.\tag{74}$$

注意到当 x 趋向 ∞ 时，$\psi_4(x)$ 必须趋向 0，所以我们取线性组合

$$\psi_4(x)=Ee^{-\mathrm{i}\alpha x}.\tag{75}$$

即

$$\begin{cases}\psi_2(x)=C_1e^{\beta x}+C_2e^{-\beta x}, & 0\le x\le a,\\ \psi_3(x)=D_1e^{\mathrm{i}kx}+D_2e^{-\mathrm{i}kx}, & a\le x\le b,\\ \psi_4(x)=Ee^{-\mathrm{i}\alpha x}, & x\ge b.\end{cases}\tag{76}$$

我们要求 $\psi(x)$ 及其一阶微商在边界处连续，所以

在 $x=0$ 处：

$$\psi_1(0)=0=\psi_2(0)=C_1e^{\beta 0}+C_2e^{-\beta 0}. \tag{77}$$

得到

$$C_1+C_2=0, \qquad C_1=-C_2. \tag{78}$$

在 $x=a$ 处：$\psi_2(a)=\psi_3(a)$，$\psi_2'(a)=\psi_3'(a)$，并利用 (78) 式，得到

$$C_1e^{\beta a}-C_1e^{-\beta a}=D_1e^{\mathrm{i}ka}+D_2e^{-\mathrm{i}ka}. \tag{79}$$

$$\beta C_1e^{\beta a}+\beta C_1e^{-\beta a}=\mathrm{i}kD_1e^{\mathrm{i}ka}-\mathrm{i}kD_2e^{-\mathrm{i}ka}. \tag{80}$$

在 $x=b$ 处：$\psi_3(b)=\psi_4(b)$，$\psi_3'(b)=\psi_4'(b)$，得到

$$D_1e^{\mathrm{i}kb}+D_2e^{-\mathrm{i}kb}=Ee^{-\mathrm{i}\alpha b}. \tag{81}$$

$$\mathrm{i}kD_1e^{\mathrm{i}kb}-\mathrm{i}kD_2e^{-\mathrm{i}kb}=-\mathrm{i}\alpha Ee^{-\mathrm{i}\alpha b}. \tag{82}$$

(79) 和 (80) 式合并同类项

$$C_1(e^{\beta a}-e^{-\beta a})=D_1e^{\mathrm{i}ka}+D_2e^{-\mathrm{i}ka}. \tag{83}$$

$$\beta C_1(e^{\beta a}+e^{-\beta a})=\mathrm{i}k(D_1e^{\mathrm{i}ka}-D_2e^{-\mathrm{i}ka}). \tag{84}$$

两式相除

$$\beta\frac{e^{\beta a}+e^{-\beta a}}{e^{\beta a}-e^{-\beta a}}=\mathrm{i}k\frac{D_1e^{\mathrm{i}ka}-D_2e^{-\mathrm{i}ka}}{D_1e^{\mathrm{i}ka}+D_2e^{-\mathrm{i}ka}} \tag{85}$$

另外，考虑到

$$\cosh\theta=\frac{e^{\theta}+e^{-\theta}}{2}, \qquad \sinh\theta=\frac{e^{\theta}-e^{-\theta}}{2}. \tag{86}$$

易得

$$\coth\theta=\frac{\cosh\theta}{\sinh\theta}=\frac{e^{\theta}+e^{-\theta}}{e^{\theta}-e^{-\theta}}. \tag{87}$$

上述 θ 均为任意角度. 所以 (85) 式可化为

$$\beta\coth\beta a=\mathrm{i}k\frac{D_1e^{\mathrm{i}ka}-D_2e^{-\mathrm{i}ka}}{D_1e^{\mathrm{i}ka}+D_2e^{-\mathrm{i}ka}}. \tag{88}$$

另外，我们再让 (81) 和 (82) 两式相除

$$\alpha=-k\frac{D_1e^{\mathrm{i}kb}-D_2e^{-\mathrm{i}kb}}{D_1e^{\mathrm{i}kb}+D_2e^{-\mathrm{i}kb}} \tag{89}$$

现在，让我们将 (88) 和 (89) 式的分式上下同除以 D_1，并令 $\frac{D_2}{D_1}=F$，则可化为

$$\beta\coth\beta a=\mathrm{i}k\frac{e^{\mathrm{i}ka}-\frac{D_2}{D_1}e^{-\mathrm{i}ka}}{e^{\mathrm{i}ka}+\frac{D_2}{D_1}e^{-\mathrm{i}ka}}=\mathrm{i}k\frac{e^{\mathrm{i}ka}-Fe^{-\mathrm{i}ka}}{e^{\mathrm{i}ka}+Fe^{-\mathrm{i}ka}}. \tag{90}$$

$$\alpha = -k\frac{e^{\mathrm{i}kb} - \frac{D_2}{D_1}e^{-\mathrm{i}kb}}{e^{\mathrm{i}kb} + \frac{D_2}{D_1}e^{-\mathrm{i}kb}} = -k\frac{e^{\mathrm{i}kb} - Fe^{-\mathrm{i}kb}}{e^{\mathrm{i}kb} + Fe^{-\mathrm{i}kb}}. \tag{91}$$

对 (91) 式展开求 F:

$$\alpha e^{\mathrm{i}kb} + \alpha F e^{-\mathrm{i}kb} = -ke^{\mathrm{i}kb} + kFe^{-\mathrm{i}kb}. \tag{92}$$

$$F = \frac{(\alpha + k)e^{\mathrm{i}kb}}{(k - \alpha)e^{-\mathrm{i}kb}}. \tag{93}$$

对 (90) 式的 F 合并同类项:

$$[\beta \coth \beta a + \mathrm{i}k]Fe^{-\mathrm{i}ka} = [\mathrm{i}k - \beta \coth \beta a]e^{\mathrm{i}ka}. \tag{94}$$

将 (93) 式代入 (94) 式, 并移项

$$[\beta \coth \beta a + \mathrm{i}k]\frac{(\alpha + k)e^{\mathrm{i}kb}}{(k - \alpha)e^{-\mathrm{i}kb}}e^{-\mathrm{i}ka} = [\mathrm{i}k - \beta \coth \beta a]e^{\mathrm{i}ka}. \tag{95}$$

$$[\beta \coth \beta a + \mathrm{i}k](\alpha + k)e^{\mathrm{i}k(b-a)} = [\mathrm{i}k - \beta \coth \beta a](k - \alpha)e^{-\mathrm{i}k(b-a)}. \tag{96}$$

我们将上式展开:

$$\begin{aligned} &\beta \coth(\beta a)\alpha e^{\mathrm{i}k(b-a)} + \mathrm{i}k\alpha e^{\mathrm{i}k(b-a)} + \beta \coth(\beta a)ke^{\mathrm{i}k(b-a)} + \mathrm{i}k^2 e^{\mathrm{i}k(b-a)} \\ &= \mathrm{i}k^2 e^{-\mathrm{i}k(b-a)} - \beta \coth(\beta a)ke^{-\mathrm{i}k(b-a)} - \mathrm{i}k\alpha e^{-\mathrm{i}k(b-a)} + \beta\alpha \coth(\beta a)e^{-\mathrm{i}k(b-a)}. \end{aligned} \tag{97}$$

并再次合并同类项

$$\begin{aligned} &\beta\alpha \coth(\beta a)[e^{\mathrm{i}k(b-a)} - e^{-\mathrm{i}k(b-a)}] + \mathrm{i}k^2[e^{\mathrm{i}k(b-a)} - e^{-\mathrm{i}k(b-a)}] \\ &= -\mathrm{i}k\alpha[e^{\mathrm{i}k(b-a)} + e^{-\mathrm{i}k(b-a)}] - \beta k \coth(\beta a)[e^{\mathrm{i}k(b-a)} + e^{-\mathrm{i}k(b-a)}]. \end{aligned} \tag{98}$$

考虑到

$$\tan\theta = -\mathrm{i}\frac{e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta}}{e^{\mathrm{i}\theta} + e^{-\mathrm{i}\theta}}. \tag{99}$$

我们让 (98) 式两边同除 $e^{\mathrm{i}k(b-a)} + e^{-\mathrm{i}k(b-a)}$ 则

$$\beta\alpha \coth(\beta a)\frac{e^{\mathrm{i}k(b-a)} - e^{-\mathrm{i}k(b-a)}}{e^{\mathrm{i}k(b-a)} + e^{-\mathrm{i}k(b-a)}} + \mathrm{i}k^2\frac{e^{\mathrm{i}k(b-a)} - e^{-\mathrm{i}k(b-a)}}{e^{\mathrm{i}k(b-a)} + e^{-\mathrm{i}k(b-a)}} = -\mathrm{i}k\alpha - \beta k \coth(\beta a). \tag{100}$$

或者

$$-\beta\alpha \coth(\beta a)\frac{e^{\mathrm{i}k(b-a)} - e^{-\mathrm{i}k(b-a)}}{e^{\mathrm{i}k(b-a)} + e^{-\mathrm{i}k(b-a)}} + k^2 \tan k(b-a) = \mathrm{i}k\alpha + \beta k \coth(\beta a). \tag{101}$$

(101) 式两边同乘 i，得

$$-\mathrm{i}\beta\alpha \coth(\beta a)\frac{e^{\mathrm{i}k(b-a)} - e^{-\mathrm{i}k(b-a)}}{e^{\mathrm{i}k(b-a)} + e^{-\mathrm{i}k(b-a)}} + \mathrm{i}k^2 \tan k(b-a) = -k\alpha + \mathrm{i}\beta k \coth(\beta a). \tag{102}$$

整理可得

$$\begin{aligned} \tan k(b-a) &= \frac{-k\alpha + \mathrm{i}\beta k \coth(\beta a)}{\mathrm{i}k^2 + \beta\alpha \coth(\beta a)} = \frac{-\mathrm{i}k\alpha - \beta k \coth(\beta a)}{-k^2 + \mathrm{i}\beta\alpha \coth(\beta a)} \\ &= \frac{\beta k \coth(\beta a) + \mathrm{i}k\alpha}{k^2 - \mathrm{i}\beta\alpha \coth(\beta a)}. \end{aligned} \tag{103}$$

除了上述解法，我们还可将 (77) 至 (82) 式联立构成系数矩阵，然后令矩阵的系数行列式为 0，也可得到 (103) 式的结果 (详见学习指导书 34 页).